Unit: 6 Unit 1: Ratio, Rate, Percentage **Name:** ________________
Topic: Ratios Proportions **Date:** ____________

Advanced (Habanero)

Q1. Which of the following represents the ratio 3 : 5?
(A) 3/5
(B) 5/3
(C) $3 \cdot 5$
(D) $5 \cdot 3$
(E) $3 + 5$

Q2. Solve the proportion $\frac{4}{6} = \frac{x}{9}$ using cross multiplication.
(A) $x = 4$
(B) $x = 6$
(C) $x = 5$
(D) $x = 3$
(E) $x = 6$

Q3. A recipe for making cookies requires a ratio of 2 : 3 for sugar and flour. If 6 cups of flour are used, how many cups of sugar are needed?
(A) 2
(B) 3
(C) 4
(D) 6
(E) 8

Q4. A map has a scale of 1 : 1000. If a distance on the map is 5 cm, what is the actual distance it represents?
(A) 5 m
(B) 50 m
(C) 500 m
(D) 5000 m
(E) 50000 m

Q5. A farmer has 48 acres of land and wants to plant a ratio of 3 : 5 of wheat to corn. How many acres will be planted with wheat?
(A) 16
(B) 18
(C) 20
(D) 24
(E) 30

Q6. Solve the proportion $\frac{2}{5} = \frac{8}{x}$ using cross multiplication.
(A) $x = 15$
(B) $x = 20$
(C) $x = 16$
(D) $x = 10$
(E) $x = 25$

Q7. A group of friends want to share some candy in a ratio of 2 : 3 : 5. If they have 50 pieces of candy, how many will each person get?
(A) $5, 5, 10$
(B) $10, 15, 25$
(C) $6, 9, 15$
(D) $8, 12, 20$
(E) $4, 6, 10$

Q8. A bicycle travels 25 miles in 5 hours. How many miles will it travel in 8 hours, assuming the same rate?
(A) 20
(B) 25
(C) 30
(D) 40
(E) 50

Open Ended Questions (FRQ)

Q9. A bakery is making a special batch of cookies for a holiday sale. The recipe calls for a ratio of 2 : 3 : 5 for flour, sugar, and chocolate chips. If the bakery has 40 pounds of flour, 60 pounds of sugar, and 80 pounds of chocolate chips, how many batches of cookies can they make? Show your work and explain your reasoning.

Q10. A car travels from City A to City B at an average speed of 60 miles per hour. On the way back, the car travels at an average speed of 40 miles per hour. What is the average speed for the entire round trip? Use a proportion to solve the problem and show your work.

Chef's Tips

- Emphasize the importance of simplifying ratios and proportions.
- Use real-world examples to illustrate the concepts of ratios and proportions.
- Encourage students to check their work by plugging in their answers to the original equation.